```
    "logger.org.elasticsearch.http":"TRACE"
  }
}
```

This code sets the `DEBUG` level on the `org.elasticsearch.index` package and the `TRACE` level on the `org.elasticsearch.http` package. Because both are *persistent* properties, the logger writes detailed logs at these levels as set on the packages, the properties survive cluster restarts (or crashes).

Be careful about setting such properties permanently using the `persistent` property. I suggest enabling the `DEBUG` or `TRACE` logging level when troubleshooting or during a debugging episode. When you finish dealing with the episode in production, reset the level to `INFO` to avoid writing reams of requests to disk.

14.5.3 Java virtual machine options

Because Elasticsearch uses the Java programming language, many optimization tweaks can be done at the Java virtual machine (JVM) level. For obvious reasons, discussing such a huge topic in this book would not do it justice. However, if you are curious and want to understand the JVM or fine-tune performance at a lower level, refer to a book like *Optimizing Java* (Ben Evan and Jame Gough, 2018, O'Reilly) or *Java Performance* (Scott Oaks, 2020, O'Reilly). I highly recommend them as they provide the fundamentals along with operational tips and tricks.

Elasticsearch provides a jvm.options file in the /config directory with the JVM settings. However, this file is only for reference purposes (to check the node's memory settings, for example) and never to be edited. Heap memory is automatically set for the Elasticsearch server based on the node's available memory.

> **WARNING** Never edit the jvm.options file under any circumstances. Doing so may corrupt Elasticsearch's internal workings.

To upgrade memory or change JVM settings, we must create a new file with .options as the filename extension, provide the appropriate tuning parameters, and place the file in a directory called jvm.options.d under the config folder for archive installations (tar or zip). We can give any name to the custom file, but we must include the fixed .options extension.

For RPM/Debian package installations, this file should be under the /etc/elasticsearch/jvm.options.d/ directory. Similarly, mount the options file under the /usr/share/elasticsearch/config folder for Docker installations.

We can edit the settings in this custom JVM options file. For example, to upgrade heap memory in a file called jvm_custom.options, we can use the following code.

Listing 14.16 Upgrading heap memory

```
-Xms4g
-Xmx8g
```